27 March 2023
Department of Computer Science

VRIJE UNIVERSITEIT AMSTERDAM

Final Exam

Logic and Sets for AI

This exam has 2 pages and 5 exercises. The result will be computed as $1 + \left(\frac{total}{68} \cdot 9\right)$

Note that in this course, 0 is considered a natural number. If you are, in any exercise, unsure about the definition of some term/operator/... state your assumed definition at the beginning of the exercise.

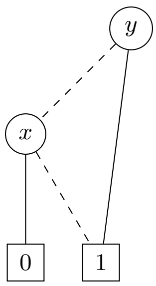
1. OBDDs (2 + 6 + 2 + 6 points)

(a) Represent the formula $\phi = (x \leftrightarrow y) \vee (z \wedge x)$ by a binary decision tree against the variable order y, z, x.

(b) Transform the binary decision tree from (a) into a reduced OBDD. Show **all** the steps and give the **name** for each step.

(c) Briefly explain how to construct an OBDD for $O_1 \rightarrow O_2$ from two reduced OBDDs O_1 and O_2 against the same variable order for arbitrary O_1 and O_2. (Hint: You can refer to content/algorithms shown in the lectures).

The formula ψ is represented by the following OBDD:



(d) Use the process from (c) to construct an OBDD for $\phi \rightarrow \psi$ from the reduced OBDDs for ϕ (constructed in (b)) and ψ (see above) and transform it into a reduced OBDD. Again, show **all** the steps and give the **name** for each step of the reduction.

2. Predicate Logic (6 + 8 points)

For each of the following pairs of formulas, either argue that they are semantically equivalent, or give a model on which their truth values are different with a brief explanation. If they are not equivalent, argue whether one entails the other or give a counterexample.

(a) $\exists x \forall y\, R(y, x)$ and $\forall y \exists x\, R(x, y)$

(b) $\exists y \forall x \big(R(x, y) \oplus R(y, x)\big)$ and $\forall x \exists y \big(R(x, y) \oplus R(y, x)\big)$

3. Equivalence Relations (6 + 5 + 3 points)

The relation R is defined on the set A = {2, 4, 6, 8} × {2, 4, 6, 8} in the following way:

$$\langle a, b\rangle \ R \ \langle c, d\rangle \qquad \Longleftrightarrow \qquad a \cdot b = c \cdot d$$

(a) Show that R is an equivalence relation.

(b) Explicitly list all of the equivalence classes of R on the set A.

(c) Give a complete system of representatives for these equivalence classes.

4. Functions (2 + 4 + 8 points)

Consider the function *div*: $\mathbb{R} \to \mathbb{R}$ defined by

$$div\,(x) = \frac{1}{x}$$

(a) Give the **domain of definition** and **range** for the function *div*. Briefly motivate your answers.

(b) Explain whether or not the function is **injective** and whether or not it is **surjective**.

Now also consider the functions *incr*: $\mathbb{N} \to \mathbb{N}$ and *tr*: $\mathbb{R} \to \mathbb{Z}$ defined by

$$incr(x) = x + 1 \qquad\qquad tr(x) = [x]$$

where $[x]$ here means truncating x to its integer part, i.e., $[4.2] = 4$ and $[-6.9] = -6$.

The composite function f is defined by

$$f := tr \circ div \circ incr$$

(c) Explain whether the function f is **total** and determine its **range**. Hint: A Venn-Diagram of the composite function might help visualize your explanation.

5. Induction (2 + 8 points)

We claim that the following statement is true for all natural numbers $n \geq 1$:

$$\sum_{k=1}^{n} k! \cdot k = (n + 1)! - 1$$

(recall that $n!$ is a factorial function which is defined as $n! = 1 \cdot 2 \cdot 3 \cdot \ldots \cdot n$)

(a) Verify by explicit computation that the claim is true for $n = 1$ and $n = 2$.

(b) Prove by mathematical induction that the statement holds true for all natural numbers $n \geq 1$.